

Blake McAlister

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EDUCATION

Clemson University

BS in Computer Science, Minor in Artificial Intelligence (GPA: 3.3/4.0)

Clemson, SC

Graduation Date: December 2026

SKILLS

Programming: Python, Java, C/C++, JavaScript, OOP, Data Structures & Algorithms, Performance Optimization

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Computer Vision (YOLO, OpenCV, OCR), NLP, Supervised & Unsupervised Learning, Model Training & Evaluation (Precision, Recall, F1, AUC), Vision-Language Models

Data Science & Analytics: NumPy, Pandas, Data Cleaning, Exploratory Data Analysis, Statistical Analysis, Data Visualization

Web Development: Flask, FastAPI, React, REST APIs, Backend Development, API Integration, Microservices Architecture

Systems: Git/GitHub, Linux/Unix, High-Performance Computing (GPU, Palmetto Cluster), Unit & Integration Testing

EXPERIENCE

Watt AI Creative Inquiry Research

Clemson, SC

Audiogram Deep Learning Researcher

August 2025 – December 2025

- Built a robust end-to-end ML pipeline integrating YOLO detection, coordinate calibration, and preprocessing to digitize handwritten clinical audiograms with high reliability.
- Engineered advanced OCR + Vision-LLM extraction systems to automatically interpret speech scores and output structured, Health Record ready data.
- Led deployment and documentation of framework, using Python, PyTorch/TensorFlow, OpenCV, Git, and HPC workflows.

Oldcastle APG

Atlanta, GA

Digital Solutions Intern

May 2025 – August 2025

- Integrated a chatbot into Oldcastle's B2B e-commerce platform to reduce manual support load by automating order tracking, FAQs, and customer support workflows.
- Assisted with backend improvements and API integrations within the BigCommerce ecosystem to support scalable commerce operations.
- Analyzed user behavior using Lucky Orange to identify friction points, improve site performance, and guide UX and UI.

CodePath x AmazonNext Program

Atlanta, GA

Software Engineering Fellow

June 2024 – August 2024

- Selected for an industry-backed program emphasizing core SWE fundamentals and technical interview readiness.
- Applied data structures and algorithmic concepts through hands-on coding challenges and problem-solving exercises.
- Built proficiency in Big O analysis, hash tables, linked lists, stacks and queues, binary trees, graphs, strings, and arrays.

RELEVANT PROJECTS:

Urba

October 2025

- Built a real-time environmental alert platform at the NASA Space Apps Hackathon using NASA EONET and public APIs.
- Implemented a social-feed system for live mobile app, location-based reporting of fires, air quality, and extreme weather.
- Engineered a city-level Wellness Score covering U.S. cities by aggregating live environmental and user-reported data.

Vision-Based Autonomous RC Car

November 2025

- Implemented real-time lane detection + heading correction using computer vision on a TurboPi system.
- Added car-following logic: distance estimation, stopping behavior, and restart logic for safe trailing.
- Achieved stable tracking in real hallway tests and multi-lane simulation environments.

Stock Price Predictor

January 2025

- A web-based application using Streamlit to predict stock prices, leveraging Python's data science and machine learning libraries. Implemented real-time data retrieval from the Yahoo Finance API, visual moving averages (100, 200, and 250 days)

EXTRACURRICULARS

ColorStack(Vice President). National Society of Black Engineers (NSBE). AI Club. Clemson WBB Practice Player. Club Basketball. Dae Han Black Belt.

Study Abroad: Mexico (May 2024): Merida, Campeche, Chichen Itza, Valladolid, Puerto Morelos.